

high and remains high for one second. Transistor T1 inverts the output of IC2 to make input pin 2 of IC3 low.

As soon as IC2 is triggered, the leading edge of the positive-going output pulse sends resting signal to all the decade counters via capacitor C6. The five 7-segment displays (DIS1 through DIS5) now show the count as '00000.' As pin 2 of IC3 is low, the counter is enabled to count for one second. It continues counting the input pulses as long as the output of IC2 remains high.

After one second, the output of IC2 becomes low again. Transistor T1 is cut-off, prohibiting further counting by pulling up pin 2 of IC3 to logic '1.' The number of pulses counted so far is displayed over 7-segment displays.

The process repeats, as long as power is applied to the counter circuit.

Pin 5 of timer NE555 is the control voltage pin that is primarily used for filtering when the timer is used in noisy environments. However, by imposing a voltage at this pin, it is possible to vary the calculated period. A 0.01µF capacitor connected to pin 5 of NE555 bypasses any noise from altering the calculated pulse-width. LED2 indicates the counting period.

Fig. 2 shows the power supply circuit. The 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 9V, 500mA. The transformer output is rectified by full-wave rectifier BR1, filtered by capacitor C7 and regulated by IC 7805 (IC8). The regulated 5V DC so obtained is further filtered by capacitors C8 and C9. LED3 acts as the power indicator. Resistor R14 acts as the current limiter. A capacitor above

10 µF is connected across the output of the regulator IC, while diode D1 protects the regulator IC in case its input is shorted to ground.

Construction

Assemble the circuit on a printed circuit board (PCB) to minimise time and assembly errors. An actual-size, single-side PCB for the frequency counter is shown in Fig. 3 and its component layout in Fig. 4.

Carefully assemble the components and double-check for any overlooked error. Use a low-capacitance cable to feed signal from the oscillator source to the frequency counter. Connect a known-frequency source (like calibration signal of an oscilloscope) to the input of the counter and adjust trimpot VR1 to display the frequency on the 7-segment displays. ●



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